# REDCap Newborn Follow-up Dashboard

Standard Operating Procedure (SOP) & Technical Reference Manual — Version 3.0

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| **Document Identifier:** SOP-NHRP-DASHBOARD-2026-V3.0 | **Effective Date:** September 2026 |
| **Target Audience:** Data Analysts, Telecallers, District Supervisors, State Coordinators | **Operational Status:** Live Production Release |

## 1. Executive Overview & System Architecture

This Standard Operating Procedure (SOP V3.0) defines the end-to-end data pipeline, column mappings, data cleansing algorithms, 7-step chronological cross-dataset matching criteria, and data validation rules for the **REDCap Newborn Follow-up Dashboard**. The system aggregates birth notification, registration, and telephonic follow-up data (60,000+ records) from REDCap alongside state ANMOL delivery reports and institutional SNCU admission datasets, providing real-time KPI monitoring, 1-click line-list filtering, and automated discrepancy reporting.

| Stage | Data Source / Component | Description & Purpose |
|---|---|---|
| **1. Ingestion** | REDCap, ANMOL, & SNCU Files | Loads raw REDCap dataset (`NHRP update.xlsx`), ANMOL delivery line-lists, and SNCU admissions using Calamine engine. |
| **2. Standardization** | Data Cleansing Modules | Parses multi-format dates, swaps inverted day-month entries, standardizes blocks/villages, and validates 10-digit phones. |
| **3. Spatial Mapping** | Location & Staff Directories | Maps Block (Col AF) and Village to Subcentre, attaching ANM, CHO, and Telecaller coordinator metadata. |
| **4. ANMOL Match** | 4-Tier Matching Engine | Cross-matches ANMOL home and out-of-district births with REDCap records via MPID, phone, and fuzzy name similarity. |
| **5. SNCU Match** | 7-Step Chronological Engine | Cross-verifies SNCU neonatal deaths (<=28 days) with REDCap across 7 sequential criteria with dual record IDs. |
| **6. Validation** | Form 1 & Form 2 Rules | Evaluates 17 Form 1 checks (excluding pending cases in 1.9) and 7 Form 2 checks (evaluating Col XU/AAD/ACM). |
| **7. Standalone Build** | HTML Generator (`generate_dashboard.py`) | Bundles compressed JSON payload, Chart.js v4, and interactive UI into a self-contained offline HTML file. |

## 2. Input Data Files & Column Mapping Reference

The dashboard generator integrates multiple heterogeneous datasets. Below are the exact Excel column headers, Excel column numbers, and operational roles used in the current version:

| Category | Column No. & Exact Header Name | Variable Description & Usage |
|---|---|---|
| **Record Key** | `Record ID` | Primary key unique to each REDCap registration record. |

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| **Block Hierarchy** | **Col AF:** `Block / Taluk Name ` | Administrative health block used for spatial filtering and match criteria Step 1. |
| **Delivery Place** | **Col BY:** `Q43. If health facility, Delivery Place name (Public)23` | Public health facility name used for place filtering and match criteria Step 2. |
| **Telecallers** | `Q10. TeleCaller/Data Collector ID` **Col 639/700/761:** `Q10. TeleCaller ID410 / ID471 / ID532` | Assigned telecaller coordinators across registration and follow-up calls 1–3 (23 active coordinators). |
| **Delivery Dates** | `Q33. Delivery Date` **Col XO/ZX/ACG:** `Q9. Call Date_Time 409/470/531` | Delivery date and follow-up completion timestamps used for activity reporting. |
| **Follow-up Status** | **Col XU/AAD/ACM:** `Q15. Baby Current Status 415 / 476 / 537` | Final follow-up outcome status (Healthy, Dead, Sick, Completed) for Baby 1, 2, and 3. |
| **Mortality Data** | `Q51 / Q57 / Q63`: Date of death `Q16. If baby died, Date of death...` `Q18` series: Place of death | Neonatal death dates, place of death (Hospital/Home/Way), and age at death (<= 28 days). |

## 3. 7-Step Chronological SNCU Neonatal Death Matching Engine

To cross-verify hospital-based neonatal mortality (SNCU DH Khargone) with community REDCap records, the pipeline applies a strict chronological 7-step criteria hierarchy. Expired cases with 0 ≤ Age at Death ≤ 28 days (calculated from Date of Birth to Discharge Date) are evaluated:

| Step No. | Matching Field / Variable | REDCap Source Field | SNCU Admission Field |
|---|---|---|---|
| **Step 1** | Block / Taluk Name | **Col AF:** `Block / Taluk Name ` | `Block Name` |
| **Step 2** | Place of Delivery | **Col BY:** `Q43. Delivery Place name...` | `Place of Delivery` |
| **Step 3** | Delivery Date vs Date of Birth | `Q33. Delivery Date` | `Date of Birth` |
| **Step 4** | Name of Mother & Father | `Q2. Mother Name` / `Q3. Father Name` | `Mother Name` / `Father/Husband Name` |
| **Step 5** | Primary Contact Number | `Q16. Family Mobile Number` | `Contact No.` |
| **Step 6** | Date of Death vs Discharge Date | `Q51/Q57/Q63/Q16` (Date of Death) | `Discharge Date` (Expired date) |
| **Step 7** | Secondary Mobile Verification | `Q17. Family 2nd Mobile Number` | `Contact No.` |

- **Accurately Recorded (≥ 5 Criteria Matched):** Cases matching on 5 or more criteria out of the 7 chronological steps.
- **Recorded with Spelling Mismatch (< 5 Criteria Matched):** Cases established via DOB + Delivery Place with minor spelling differences in names/phone numbers.
- **Not Recorded in REDCap:** Cases where no corresponding delivery record exists in the REDCap follow-up repository.
- **Dual Record IDs:** Side-by-side display of SNCU Admission ID and REDCap Record ID in expandable inspection drawers.

## 4. Data Validation Quality Engine (Form 1 & Form 2 Rules)

Automated data quality checks evaluate registration completeness, gestational age consistency, and follow-up outcomes across Form 1 (1.1 to 1.17) and Form 2 (2.1 to 2.7):

| Code | Validation Rule Name | Evaluation Formula & Operational Definition |
|---|---|---|

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| **1.1** | Delivery Date Missing | `Q33. Delivery Date` is NaN or blank. |
| **1.2** | Block Name Missing | `Block / Taluk Name ` (Col AF) is NaN, blank, or 'Not Found'. |
| **1.3** | Gestational Age Missing | `Q38 (EDD)`, `Q32 (LMP)`, and `Q35/Q36 (Clinical GA)` are all blank. |
| **1.4** | Gestational Age Mismatch | Variance among derived GAs > 1.5 weeks (Priority: EDD > LMP > Clinical). |
| **1.5** | Delivery Type Missing | `Q39. Delivery type` is NaN or blank. |
| **1.6** | Delivery Location Missing | `Q41. Delivery Location` is NaN or blank. |
| **1.7** | Number of Babies Delivered Missing | `Q45. Number of baby delivered` is NaN or blank. |
| **1.8** | Baby Not Registered after Notification | `Q45` ≥ 1 baby, but `Q47 (Baby-1 Status)` is blank. |
| **1.9** | Status of Baby at Birth Mismatched | Stillbirth reported in `Q47`, but baby has completed follow-up status (Healthy/Dead/Sick) or death date (excludes Pending cases). |
| **1.10** | Stillbirth GA < 28wks or Weight < 1000g | `is_stillbirth` is True AND (`gestation` < 28 weeks OR `weight` < 1000g). |
| **1.11** | Baby Gender Missing | `Q48 / Q54 / Q60 (Sex)` is blank or 'Unknown'. |
| **1.12** | Birth Weight Missing / 9999 | Birth weight is NaN, 9999, ≤ 0g, or > 8000g. |
| **1.13** | Live Birth Gestational Age < 24wks | `is_live_born` is True AND `gestation` < 24 weeks. |
| **1.14** | Intergrowth Chart GA vs BW Mismatch | Birthweight >200g outside -4SD to +4SD for GA 24–42 weeks. |
| **1.15** | Mobile Number Issues | Both phones missing OR primary phone fails 10-digit numeric check. |
| **1.16** | Duplicate Entries | Multiple rows share identical `(Mother Name, Mobile, Delivery Date)`. |
| **1.17** | Future Date Issues | `Delivery Date` > current system date. |
| **2.1** | Day 29 Status Missing / Due | Live birth ≥ 29 days old without completed status in Col XU/AAD/ACM. |
| **2.2** | Date of Death Missing | Baby reported Dead in calls, but `Date of Death` is NaN. |
| **2.3** | Place of Death Missing | Neonatal death confirmed, but `Place of Death` is Not Recorded. |
| **2.4** | Age at Death Invalid | `age_at_death` < 0 days OR `age_at_death` > 28 days. |
| **2.5** | Factors Associated with ND Missing | Neonatal death confirmed, but cause-of-death checkboxes unchecked. |
| **2.6** | Call Date Before Delivery Date | Any follow-up `call_date` < `delivery_date`. |
| **2.7** | Death Timeline Issues | Call 3 date > death date OR Call 3 status is Healthy/Sick for deceased baby. |

## 5. Interactive UI Controls & Line List Filtering

The web dashboard provides multi-layered filtering controls to support supervisory and field-level follow-up:

1. **Global Filter Bar:** Filter simultaneously by **Month**, **Block**, **Delivery Coordinator (Col 21)**, and **Follow-up Done By (Col 639/700/761)** across all tabs.
2. **1-Click Interactive KPI Cards:** Instant single-click filtering across Home Deliveries (All, Accurately Recorded, Spelling Mismatch, Unmatched), Out-of-District Deliveries, and SNCU Deaths.
3. **Place of Delivery Sub-Filters:** Dedicated delivery location dropdowns in **Neonatal Deaths Registry (Col BY)**, **Khargone SNCU Deaths**, and **Khandwa SNCU Deaths** line-lists.
4. **Inspection Drawers:** Clickable row drawers showing field-by-field verification matrices, health worker contact cards, and chronological comparison summaries.
5. **Excel CSV Export:** One-click filtered data export with automatic file naming for ground-level distribution.

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| **Prepared By:** Technical Data Analyst Team | **Approved By:** State Health Program Director |
| **Signature:** _____ | **Signature:** _____ |
| **Date:** September 2026 | **Date:** September 2026 |